

WEEK-12

IAM

1. Administrative Setup (Root User)

Step 1: Log in to the AWS Management Console as the Root User.

Step 2: Search for IAM under the "Security, Identity, & Compliance" category.

Step 3: Navigate to Users and click Create user.

The screenshot shows the AWS IAM Dashboard in a web browser. The browser's address bar displays the URL: `us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/home`. The AWS Management Console header is visible, showing the user is logged in as Archana (8481-8656-3482). The left sidebar contains the navigation menu with 'IAM' selected and 'Dashboard' as the current view. The main content area is titled 'IAM Dashboard' and includes several key sections:

- Security recommendations**: Shows two recommendations with green checkmarks:
 - Root user has MFA: Having multi-factor authentication (MFA) for the root user improves security for this account.
 - Root user has no active access keys: Using access keys attached to an IAM user instead of the root user improves security.
- IAM resources**: A table showing the count of resources in the AWS Account:

User groups	Users	Roles	Policies	Identity providers
1	2	94	31	0
- AWS Account**: Displays account details:
 - Account ID: 848186563482
 - Account Alias: [Create](#)
 - Sign-in URL for IAM users in this account: `https://848186563482.signin.aws.amazon.com/console`
- Quick Links**: Includes a link for 'My security credentials' and a description: 'Manage your access keys, multi-factor authentication (MFA) and other credentials.'

Step 4: User Details: Enter a name (e.g., S3_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.

The screenshot shows the AWS IAM 'Create user' console in the 'us-east-1' region. The breadcrumb navigation is 'IAM > Users > Create user'. On the left, a progress bar shows four steps: 'Step 3: Review and create', 'Step 4: Retrieve password', and two unlabelled steps. The main form is for 'User name' and 'Console password'. The 'User name' field contains 'CC-LAB'. Below it, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)'. The 'Provide user access to the AWS Management Console - optional' checkbox is checked. Below this, a note says: 'In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.' The 'Console password' section has two options: 'Autogenerated password' (unselected) and 'Custom password' (selected). The 'Custom password' field contains 'user@123'. Below the field, two requirements are listed: 'Must be at least 8 characters long' and 'Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' '. The 'Show password' checkbox is checked. At the bottom, there is an unchecked checkbox for 'Users must create a new password at next sign-in - Recommended' with a note: 'Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.' A blue information box at the bottom states: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. The bottom of the console shows 'Cancel' and 'Next' buttons. The footer includes 'CloudShell', 'Feedback', 'Console Mobile App', '© 2026, Amazon Web Services, Inc. or its affiliates.', 'Privacy', 'Terms', 'Cookie preferences', and system information: '29°C', '10:54 AM', '3/25/2026'.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create

aws IAM > Users > Create user

Step 3
Review and create

Step 4
Retrieve password

User name

CC-LAB

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

user@123

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' '

☒ **Show password**

☐ **Users must create a new password at next sign-in - Recommended**
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel Next

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 29°C 10:54 AM 3/25/2026

Step 5: Set Permissions: Select "Attach policies directly." Search for and select AmazonS3FullAccess.



[Alt+S]



Archana (8481-8656-3482)

Archana

IAM > Users > Create user

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1491)

Create policy

Choose one or more policies to attach to your new user.

Filter by Type



All types 

27 matches

< 1 2 > 

	Policy name 	 Type 	Attached entities 
☐	  AmazonDMSRedshiftS3Role	AWS managed	0
☒	  AmazonS3FullAccess	AWS managed	7
☐	  AmazonS3ObjectLambdaExecutionRol...	AWS managed	0
☐	  AmazonS3OutpostsFullAccess	AWS managed	0

Step 6: Review & Create: Complete the wizard. On the final page, click Download .csv file.

The screenshot shows the AWS IAM console interface for creating a new user. The browser address bar displays `us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create`. The AWS navigation bar at the top includes the AWS logo, a search bar, and the user's profile (Archana, ID: 8481-8656-3482). The breadcrumb trail indicates the path: IAM > Users > Create user.

On the left, a vertical sidebar shows the four steps of the wizard: Step 1: Specify user details, Step 2: Set permissions, Step 3: Review and create (highlighted with a blue circle), and Step 4: Retrieve password.

The main content area is titled "Review and create" and includes a sub-header: "Review your choices. After you create the user, you can view and download the autogenerated password, if enabled."

The "User details" section contains three fields:

- User name:** CC-LAB
- Console password type:** Custom password
- Require password reset:** No

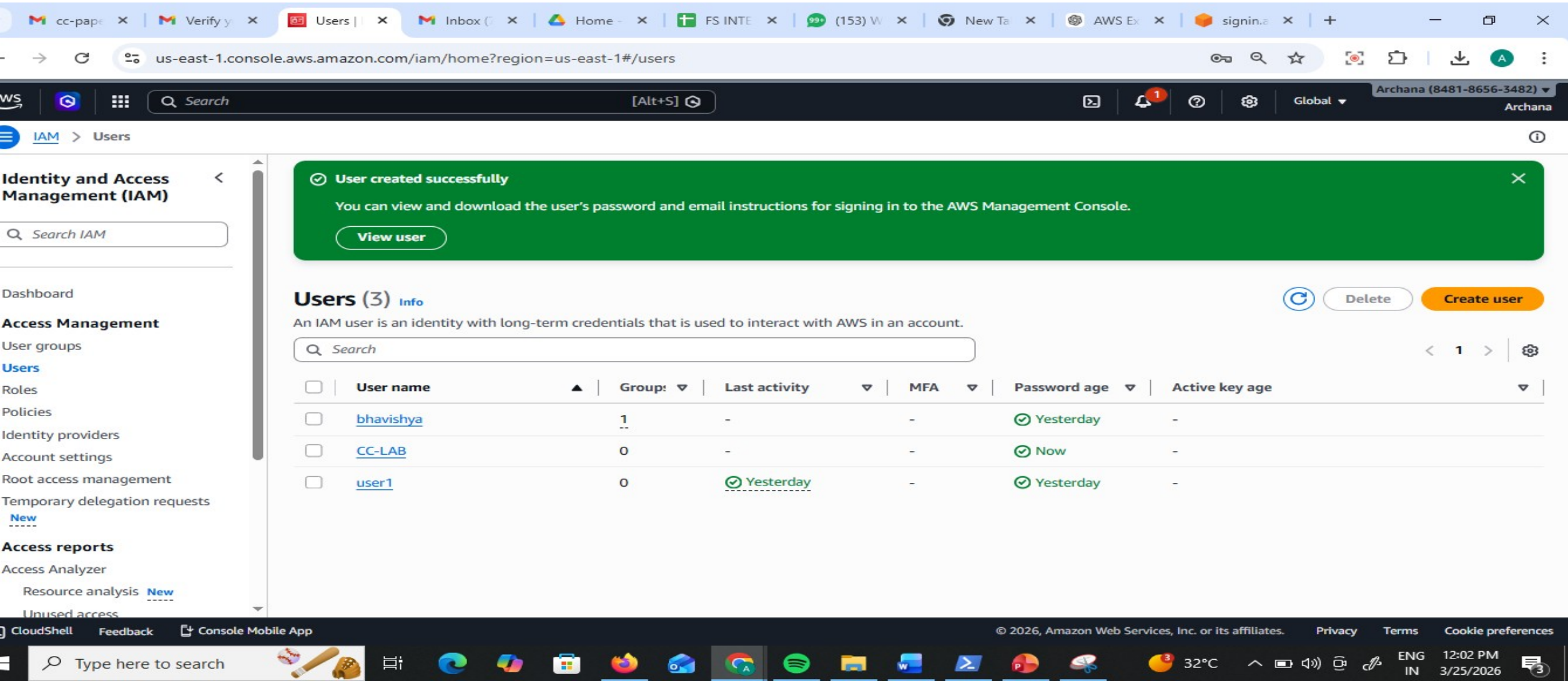
The "Permissions summary" section displays a table with one permission:

Name	Type	Used as
AmazonS3FullAccess	AWS managed	Permissions policy

The "Tags - optional" section states: "Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user. No tags associated with the resource." Below this is an "Add new tag" button and a note: "You can add up to 50 more tags."

The footer of the console shows links for CloudShell, Feedback, and Console Mobile App, along with copyright information (© 2026, Amazon Web Services, Inc. or its affiliates.) and links for Privacy, Terms, and Cookie preferences.

Now you see CC-LAB user created



The screenshot shows the AWS IAM console interface. At the top, a green banner indicates "User created successfully" with a "View user" button. Below this, the "Users (3)" section is displayed, showing a list of IAM users. The table includes columns for User name, Group, Last activity, MFA, Password age, and Active key age. The user "CC-LAB" is listed with a "Now" password age. The left sidebar shows the navigation menu with options like Dashboard, Access Management, and Access reports.

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Temporary delegation requests

[New](#)

Access reports

Access Analyzer

Resource analysis [New](#)

Unused access

Users (3) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

[Refresh](#) [Delete](#) [Create user](#)

☐	User name	Group	Last activity	MFA	Password age	Active key age
☐	bhavishya	1	-	-	✓ Yesterday	-
☐	CC-LAB	0	-	-	✓ Now	-
☐	user1	0	✓ Yesterday	-	✓ Yesterday	-

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Type here to search

32°C 12:02 PM 3/25/2026

Step 7: Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.

The screenshot shows the AWS IAM console interface. The browser address bar displays `us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users`. The console header includes the AWS logo, a search bar, and the user's name "Archana (8481-8656-3482)".

The left sidebar shows the navigation menu with "IAM" selected and "Users" in the sub-menu. The main content area displays the "Users (3)" page, which includes a search bar and a table of users.

☐	User name	Groups	Last activity	MFA
☐	bhavishya	1	-	-
☐	CC-LAB	0	-	-
☐	user1	0	✓ Yesterday	-

On the right side, the "Account ID" dropdown menu is open, showing the following information:

- Account ID: 8481-8656-3482
- Account name: Archana
- Account color: Unset
- Account settings: Organization, Service Quotas, Billing and Cost Management, Security credentials, Console Mobile App
- Buttons: "Turn on multi-session support" and "Sign out"

The bottom of the screen shows the Windows taskbar with various application icons and the system clock displaying 12:03 PM on 3/25/2026.

Step 8: Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select IAM User).

Step 9: Enter the Account ID, then the IAM Username and Password from the .csv.

The screenshot shows a web browser window with the AWS IAM user sign-in page. The browser's address bar displays the URL: `ap-southeast-2.signin.aws.amazon.com/oauth?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&code_challenge=Ne5JKSNFn4...`. The page features the AWS logo at the top center. Below the logo, there is a sign-in form titled "IAM user sign in" with a help icon. The form includes fields for "Account ID or alias" (with a link "Don't have?"), "IAM username", and "Password". The "Account ID or alias" field contains the value "848186563482". The "IAM username" field contains "CC-LAB". The "Password" field is masked with dots. There is a checkbox for "Remember this account" which is checked, and a link "Having trouble?". Below the form is an orange "Sign in" button and a link "Sign in using root user email". At the bottom of the form is a link "Create a new AWS account". To the right of the sign-in form is a large advertisement for "Database Savings Plans" with the text "Flexible pricing model that reduces your database costs by up to 35%" and a "Start saving today" link. In the bottom right corner, there is a small window titled "IAM-CLI and GUI - PowerPoint" showing a presentation slide. The Windows taskbar is visible at the bottom with various application icons and the system clock showing 12:04 PM on 3/25/2026.

Test 1 (Failure): Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied."
This proves the user is restricted.

The screenshot displays the AWS Management Console for the Asia Pacific (Sydney) region. The left-hand navigation pane shows the 'EC2' section selected, with sub-links for Dashboard, Instances, Images, and Elastic Block Store. The main content area is titled 'Resources' and lists various EC2 resources, all of which are marked with a red 'API Error' icon. The resources listed are: Instances (running) - 0, Auto Scaling Groups, Capacity Reservations, Dedicated Hosts, Elastic IPs, Instances, Key pairs, Load balancers, Placement groups, Security groups, Snapshots, and Volumes. Below the Resources section, there are three main panels: 'Launch instance' with a 'Launch instance' button, 'Service health' showing an error in retrieving service health information, and 'Account attributes' showing an error in checking for a default VPC. The right-hand sidebar shows 'EC2 cost' and 'Account attributes' sections, both of which are also marked with 'API Error' icons. The bottom of the console shows the 'Zones' section. The top of the browser window shows the URL 'ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#Home:' and the user's name 'Archana (8481-8656-3482)'.

EC2

Resources

You are using the following Amazon EC2 resources in the Asia Pacific (Sydney) Region:

Instances (running)	0	Auto Scaling Groups	API Error	Capacity Reservations	API Error
Dedicated Hosts	API Error	Elastic IPs	API Error	Instances	API Error
Key pairs	API Error	Load balancers	API Error	Placement groups	API Error
Security groups	API Error	Snapshots	API Error	Volumes	API Error

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#)

[Migrate a server](#)

Note: Your instances will launch in the Asia Pacific (Sydney) Region

Service health

[AWS Health Dashboard](#)

An error occurred
An error occurred retrieving service health information

[Diagnose with Amazon Q](#)

EC2 cost

Date range: Past 6 months | Region: Global

Total cost
Unable to load

Regions
Unable to load

[Analyze your costs in Cost Explorer](#)

Account attributes

An error occurred
An error occurred checking for a default VPC

[Diagnose with Amazon Q](#)

Settings

Zones

Test 2 (Success): Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1

aws [Search] [Alt+S] United States (N. Virginia) Archana (8481-8656-3482) CC-LAB

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

iam-user-cc-lab-bucket-demd

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

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Type here to search 32°C 12:13 PM 3/25/2026

✔ Successfully created bucket "iam-user-cc-lab-bucket-demo"

iam-user-cc-lab-bucket-demo [Info](#)

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (0)



Copy S3 URI

 Copy URL

Download

Open ↗

Delete

Actions

Create folder

 Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

🔍 Find objects by prefix

< 1 > 

	Name	Type	Last modified	Size	Storage class
--	------	------	---------------	------	---------------

No objects

You don't have any objects in this bucket.

Upload

Part B: CLI Access (Programmatic Access)

Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

1. Credential Generation

Step 1: While logged in as an Admin/Root, go to IAM > Users and click on your user.

Step 2: Go to the Security credentials tab. Scroll to Access keys and click Create access key.

The screenshot displays the AWS IAM console interface. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/CC-LAB?section=security_credentials`. The console header includes the AWS logo, a search bar, and the user's name 'Archana (8481-8656-3482)'. The left sidebar shows the navigation menu with 'IAM > Users > CC-LAB' selected. The main content area is titled 'Identity and Access Management (IAM)' and shows the 'Security credentials' tab for the user 'CC-LAB'. The user's details include 'Created March 25, 2026, 12:02 (UTC+05:30)' and 'Last console sign-in Today'. The 'Security credentials' section has four tabs: 'Permissions', 'Groups', 'Tags', and 'Security credentials' (selected). The 'Console sign-in' section shows the 'Console sign-in link' as `https://848186563482.signin.aws.amazon.com/console` and the 'Console password' as 'Updated 14 minutes ago (2026-03-25 12:02 GMT+5:30)'. The 'Last console sign-in' is '12 minutes ago (2026-03-25 12:04 GMT+5:30)'. The 'Multi-factor authentication (MFA) (0)' section shows 'No MFA devices. Assign an MFA device to improve the security of your AWS environment.' The 'Access keys (0)' section shows 'No access keys. As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials.' The bottom of the screen shows the Windows taskbar with various application icons and the system clock displaying '12:16 PM 3/25/2026'.

Step 3: Select Command Line Interface (CLI). Acknowledge the warning and click Next.

The screenshot shows the AWS IAM console interface for creating an access key. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/CC-LAB/create-access-key`. The console header shows the user is logged in as Archana (8481-8656-3482).

Progress Bar:

- Step 1: Access key best practices & alternatives (Selected)
- Step 2 - optional: Set description tag
- Step 3: Retrieve access keys

Access key best practices & alternatives [Info](#)

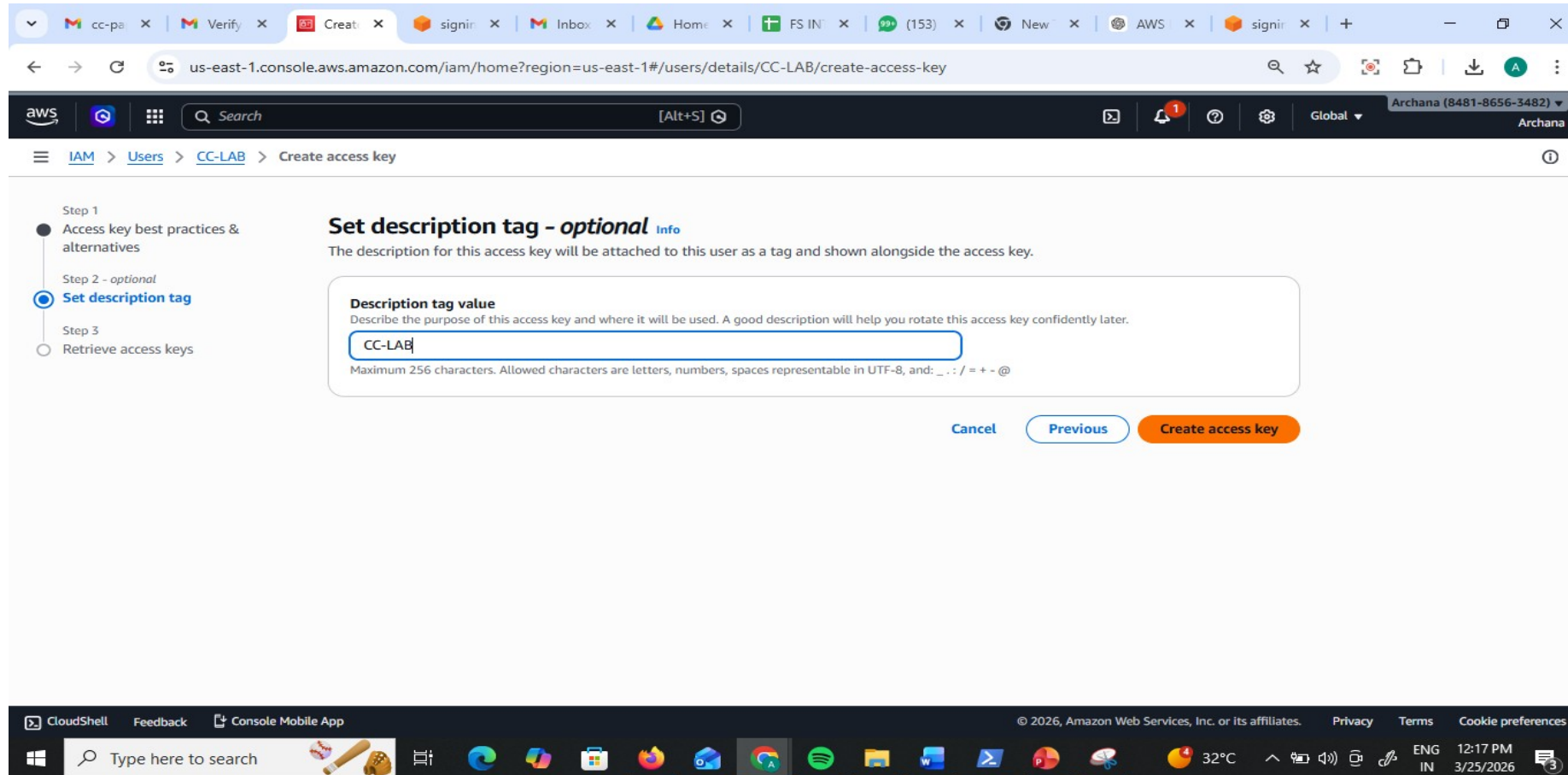
Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

- ☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.
- ☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.
- ☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
- ☐ **Third-party service**
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.
- ☐ **Application running outside AWS**
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

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Step 4: Download the .csv file. > Critical: This contains the Access Key ID and Secret Access Key. If lost, you must delete and recreate them.



✔ This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

- Step 1
Access key best practices & alternatives
- Step 2 - optional
Set description tag
- Step 3
Retrieve access keys

Retrieve access keys [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
 AKIA4K6633ONF4WWPHNT	 ***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

 CC-LAB_accessKeys.csv
99 B • Done

 Value Momentum Updated List.xlsx
37.3 KB • 27 minutes ago

 CC KR21 Question Bank (1).doc
143 KB • 2 hours ago

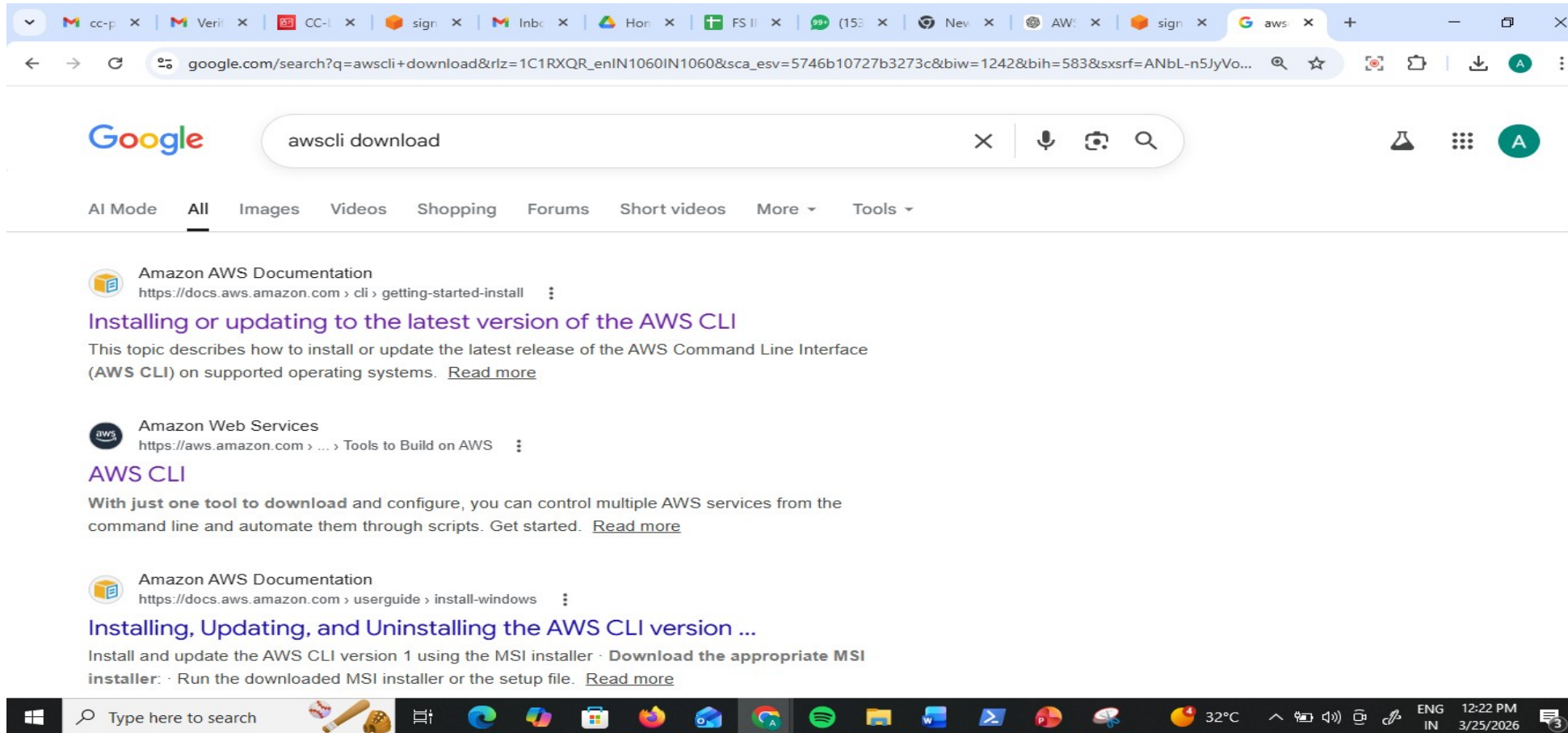
 -3482) Archana





2. Standalone System Configuration

Step 5: Install the AWS CLI Tool from google on your computer





AI resources

🔍 Search in this guide

[Return to the Console](#)

AWS Command Line Interface

User Guide for Version 2

- ▶ About the AWS CLI
- ▼ Get started
 - Prerequisites
 - Install/Update**
 - Past releases
 - Build and install from source
 - Amazon ECR Public/Docker
 - Setup
- ▶ Configure the AWS CLI
- ▶ Authentication and access credentials
- ▶ Using the AWS CLI
- ▶ AWS CLI examples
- ▶ Security
- ▶ Migration guide

For installation instructions, expand the section for your operating system.

► Linux

► macOS

► Windows

Troubleshooting AWS CLI install and uninstall errors

If you come across issues after installing or uninstalling the AWS CLI, see [Troubleshooting errors for the AWS CLI](#) for troubleshooting steps. For the most relevant troubleshooting steps, see [Command not found errors](#), [The "aws --version" command returns a different version than you installed](#), and [The "aws --version" command returns a version after uninstalling the AWS CLI](#).

On this page

AWS CLI install and update instructions

Troubleshooting AWS CLI install and uninstall errors

Next steps

▼ Recommended tasks

How to

Install specific past releases of AWS CLI version 2 on Linux

Verify Session Manager plugin installation and test access

Set up kubectl and eksctl to manage Kubernetes clusters

Configure AWS CLI to sign in with IAM Identity Center

Configure AWS CLI with IAM

Download

<https://awscli.amazonaws.com/AWSCLIV2.msi>

The screenshot shows a web browser window displaying the AWS CLI User Guide for Version 2. The browser's address bar shows the URL `docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html`. The page has a dark header with the AWS logo and navigation links: **Get started**, **Service guides**, **Developer tools**, and **AI resources**. A search bar is present with the text "Search in this guide". A yellow button labeled "Return to the Console" is in the top right.

The main content area is titled "Install and update requirements" and includes a list of requirements:

- We support the AWS CLI on Microsoft-supported versions of 64-bit Windows.
- Admin rights to install software

Below this, the section "Install or update the AWS CLI" provides instructions:

To update your current installation of AWS CLI on Windows, download a new installer each time you update to overwrite previous versions. AWS CLI is updated regularly. To see when the latest version was released, see the [AWS CLI version 2 Changelog](#) on [GitHub](#).

1. Download and run the AWS CLI MSI installer for Windows (64-bit):
<https://awscli.amazonaws.com/AWSCLIV2.msi>

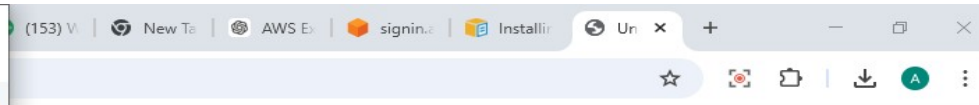
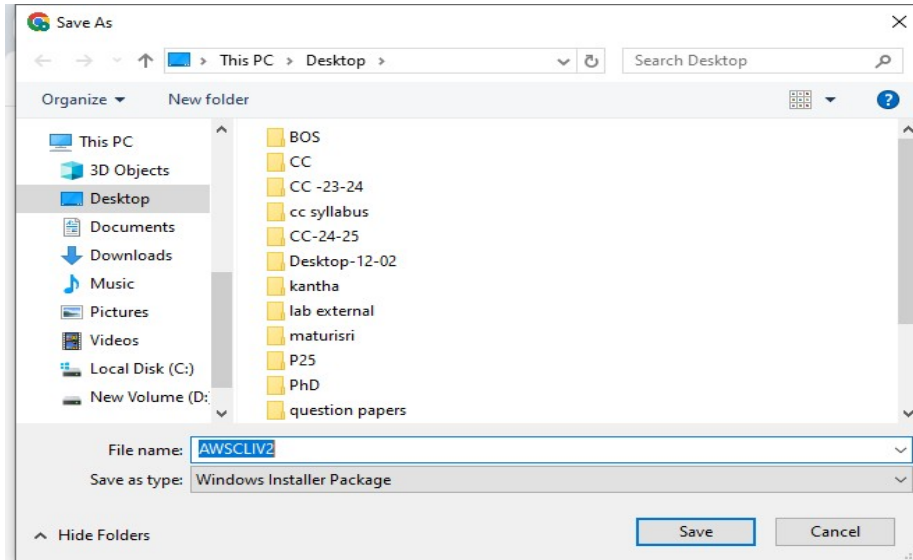
Alternatively, you can run the `msiexec` command to run the MSI installer.

```
C:\> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
```

For various parameters that can be used with `msiexec`, see [msiexec](#) on the [Microsoft Docs](#) website. For example, you can use the `/qn` flag for a silent installation.

The right sidebar contains a section "On this page" with links to "AWS CLI install and update instructions", "Troubleshooting AWS CLI install and uninstall errors", and "Next steps". Below this is a "Recommended tasks" section with links to "How to" guides, including "Install specific past releases of AWS CLI version 2 on Linux", "Verify Session Manager plugin installation and test access", "Set up kubectl and eksctl to manage Kubernetes clusters", "Configure AWS CLI to sign in with IAM Identity Center", and "Configure AWS CLI with IAM".

The bottom of the image shows a Windows taskbar with various application icons and a system tray displaying the date and time as 12:21 PM on 3/25/2026.



Step 6: Open your Command Prompt (CMD) or Terminal.
Step 7: Type aws configure and press Enter.

The image shows a web browser window with the URL `docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html`. The browser's address bar and tabs are visible at the top. Below the browser, a Windows PowerShell terminal window is open, titled "Select Windows PowerShell". The terminal displays the following text:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\De11> aws configure
AWS Access Key ID [*****QRER]:
```

The terminal window is overlaid on the AWS CLI user guide page, which is partially visible in the background. The page includes the AWS logo, navigation links, and a sidebar with a table of contents.

Step 8: Provide the following from your .csv:

Access Key ID

Secret Access Key

Default region: (e.g., ap-south-1)

Default output: json

The image shows two overlapping windows on a Windows desktop. The background window is Windows PowerShell, displaying the output of the `aws configure` command. The foreground window is Microsoft Excel, showing a spreadsheet with AWS credentials entered into cells B2 and C2.

Windows PowerShell Output:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Users\De11> aws configure
AWS Access Key ID [*****QRER]: AKIA4K6633ONF4WWPHNT
AWS Secret Access Key [*****H66J]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
```

Microsoft Excel Spreadsheet:

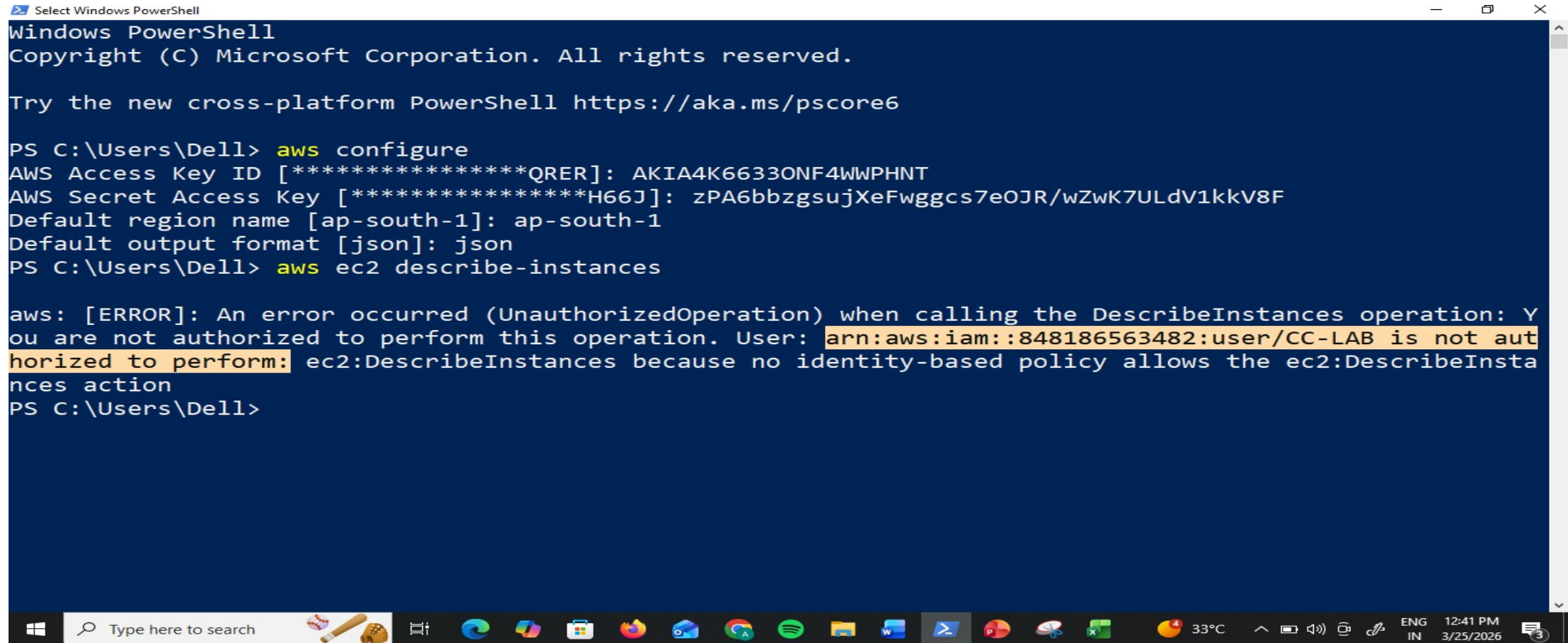
	A	B	C	D	E	F	G	H
1	Access key	Secret access key						
2	AKIA4K66	zPA6bbzgsujXeFwggcs7eOJR/wZwK7ULdV1kkV8F						
3								
4								
5								
6								
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The Excel window title is "CC-LAB_accessKeys". The status bar at the bottom shows "Ready" and "Accessibility: Unavailable". The taskbar at the very bottom includes the Windows Start button, a search bar, and various application icons. The system clock shows 12:28 PM on 3/25/2026, with a temperature of 33°C.

Run `aws ec2 describe-instances`.

Result: You will get an (AccessDenied) error.

This confirms that the CLI is respecting the same IAM policy as the GUI.



```
Select Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Dell> aws configure
AWS Access Key ID [*****QRER]: AKIA4K6633ONF4WWPHNT
AWS Secret Access Key [*****H66J]: zPA6bbzgsujXeFwggs7eOJR/wZwK7ULdV1kkV8F
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\Dell> aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:iam::848186563482:user/CC-LAB is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
PS C:\Users\Dell>
```

3. Verification via Commands

Step 9: Run `aws s3 ls`. This should list your buckets

```
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Windows PowerShell
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PS C:\Users\Dell> aws configure
AWS Access Key ID [*****QRER]: AKIA4K6633ONF4WWPHNT
AWS Secret Access Key [*****H66J]: zPA6bbzgusjXeFwggs7eOJR/wZwK7ULdV1kkV8F
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\Dell> aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:iam::848186563482:user/CC-LAB is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
PS C:\Users\Dell> aws s3 ls
2026-03-20 12:56:19 elasticbeanstalk-us-east-1-848186563482
2026-03-25 12:13:38 iam-user-cc-lab-bucket-demo
PS C:\Users\Dell>
```

Step 10: Run `aws s3 mb s3://your-unique-bucket-name`. (Success)

```
Select Windows PowerShell
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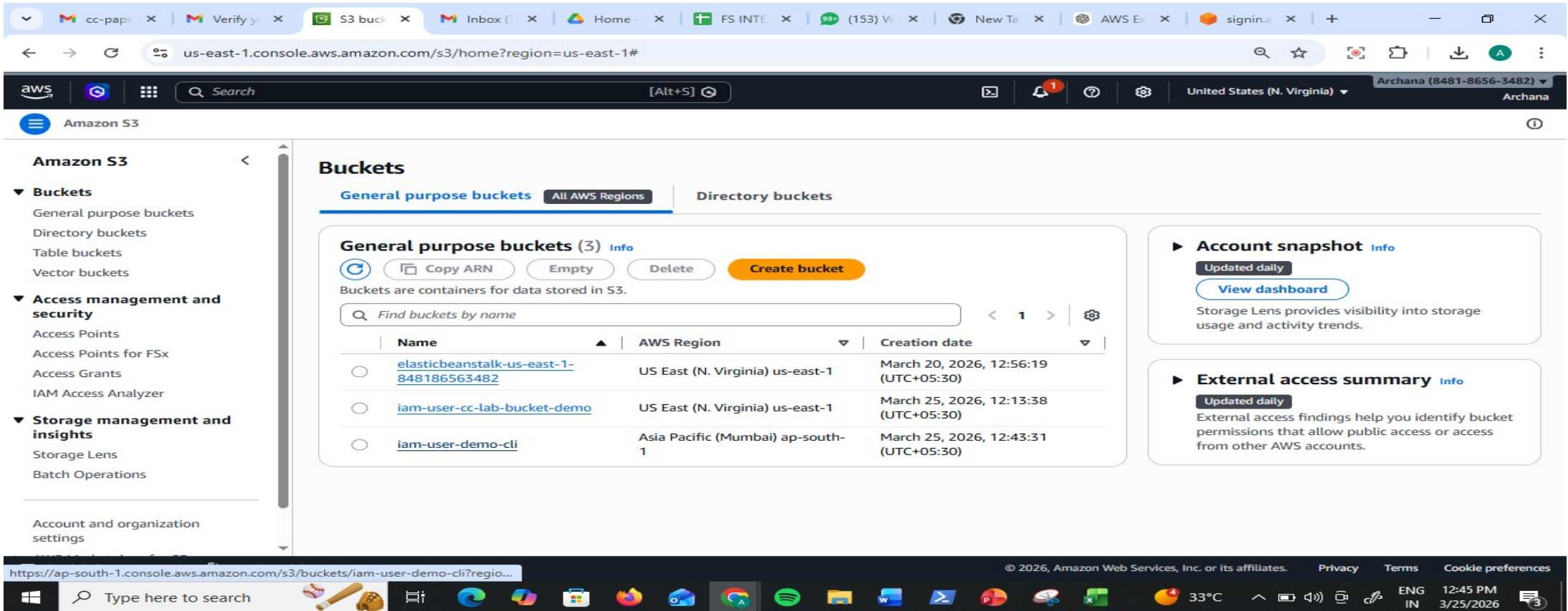
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Dell> aws configure
AWS Access Key ID [*****Q RER]: AKIA4K6633ONF4WWPHNT
AWS Secret Access Key [*****H66J]: zPA6bbzgsujXeFwggs7eOJR/wZwK7ULdV1kkV8F
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\Dell> aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform this operation. User: arn:aws:iam::848186563482:user/CC-LAB is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
PS C:\Users\Dell> aws s3 ls
2026-03-20 12:56:19 elasticbeanstalk-us-east-1-848186563482
2026-03-25 12:13:38 iam-user-cc-lab-bucket-demo
PS C:\Users\Dell> aws s3 mb iam-user-demo-cli

aws: [ERROR]: An error occurred (ParamValidation): <S3Uri>
Error: Invalid argument type
PS C:\Users\Dell> aws s3 mb s3://iam-user-demo-cli
make_bucket: iam-user-demo-cli
PS C:\Users\Dell>
```

Check in console you see the bucket created from CLI



The screenshot shows the AWS Management Console for the us-east-1 region. The left sidebar contains the navigation menu with categories like Buckets, Access management and security, and Storage management and insights. The main content area is titled 'Buckets' and shows 'General purpose buckets (3)'. A table lists the buckets:

	Name	AWS Region	Creation date
☐	elasticbeanstalk-us-east-1-848186563482	US East (N. Virginia) us-east-1	March 20, 2026, 12:56:19 (UTC+05:30)
☐	iam-user-cc-lab-bucket-demo	US East (N. Virginia) us-east-1	March 25, 2026, 12:13:38 (UTC+05:30)
☒	iam-user-demo-cli	Asia Pacific (Mumbai) ap-south-1	March 25, 2026, 12:43:31 (UTC+05:30)

Below the table, there are two informational cards: 'Account snapshot' and 'External access summary'. The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 12:45 PM on 3/25/2026.

Scenario 1: Security Risk

Q: You logged into AWS using the root user for daily work. What's wrong here?

✔ Scenario 2: Account Compromise

Q: Your AWS bill suddenly increases and you suspect root user compromise. What steps will you take?

✔ Scenario 3: Root Credentials Lost

Q: What if you forgot root user password?

✔ Scenario 4: Root Access Keys

Q: Why should you avoid creating access keys for root user?

✔ Scenario 5: New Developer Joining

Q: A new developer joins your team. How do you give access?

✔ Scenario 6: Least Privilege

Q: A user needs only S3 read access. What will you do?

✔ Scenario 7: Temporary Access

Q: A contractor needs access for 2 weeks.

✔ Scenario 8: Access Denied Error

Q: User gets “Access Denied” while accessing EC2. Why?

✔ Scenario 9: Multiple Users Same Permissions

Q: 50 users need same access. What is best practice?

✔ Scenario 10: Configure CLI

Q: You installed AWS CLI. How do you configure it?

✔ Scenario 11: Create IAM User via CLI

Q: How to create IAM user using CLI?

✔ Scenario 12: Attach Policy via CLI

Q: Give S3 read access to user using CLI

✔ Scenario 13: Create Access Keys via CLI

Q: How to generate access keys for IAM user?

✔ Scenario 14: List IAM Users

Q: How to check all users using CLI?

✔ Scenario 15: Delete IAM User via CLI

Q: How to safely delete a user?

1Q: CLI command fails with AccessDenied. What to check?

